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Abstract

We demonstrate **Exceptional point (EP)** induced control of spontaneous emission in a one-dimensional (1D) second-order grating integrated with an **InAs/GaAs quantum dot membrane**. By engineering the grating duty cycle, we realize an EP through radiative coupling with the back-propagating mode. Low-temperature angle-resolved photoluminescence measurements show **EP band coalescence** and **localized momentum-resolved emission**. Time-resolved photoluminescence further shows reduced emission lifetime at the EP, indicating enhanced radiative decay via modified local density of optical states (LDOS). Our **EP integrated quantum dot** provides a platform to explore non-Hermitian photonics for quantum light source generations.

Theoretical Model

$$H(k, \Delta) = E_0 + \begin{pmatrix} \frac{\Delta + \delta k^2}{2} & v \cdot k \\ v \cdot k & -\frac{\Delta + \delta k^2}{2} \end{pmatrix} + i \begin{pmatrix} \gamma_{nr} + \gamma_b & 0 \\ 0 & \gamma_{nr} \end{pmatrix}$$

Bright mode energy (red box) Dark mode energy (blue box) Non-Hermitian part (purple box)

Δ : Energy gap between dark and bright modes

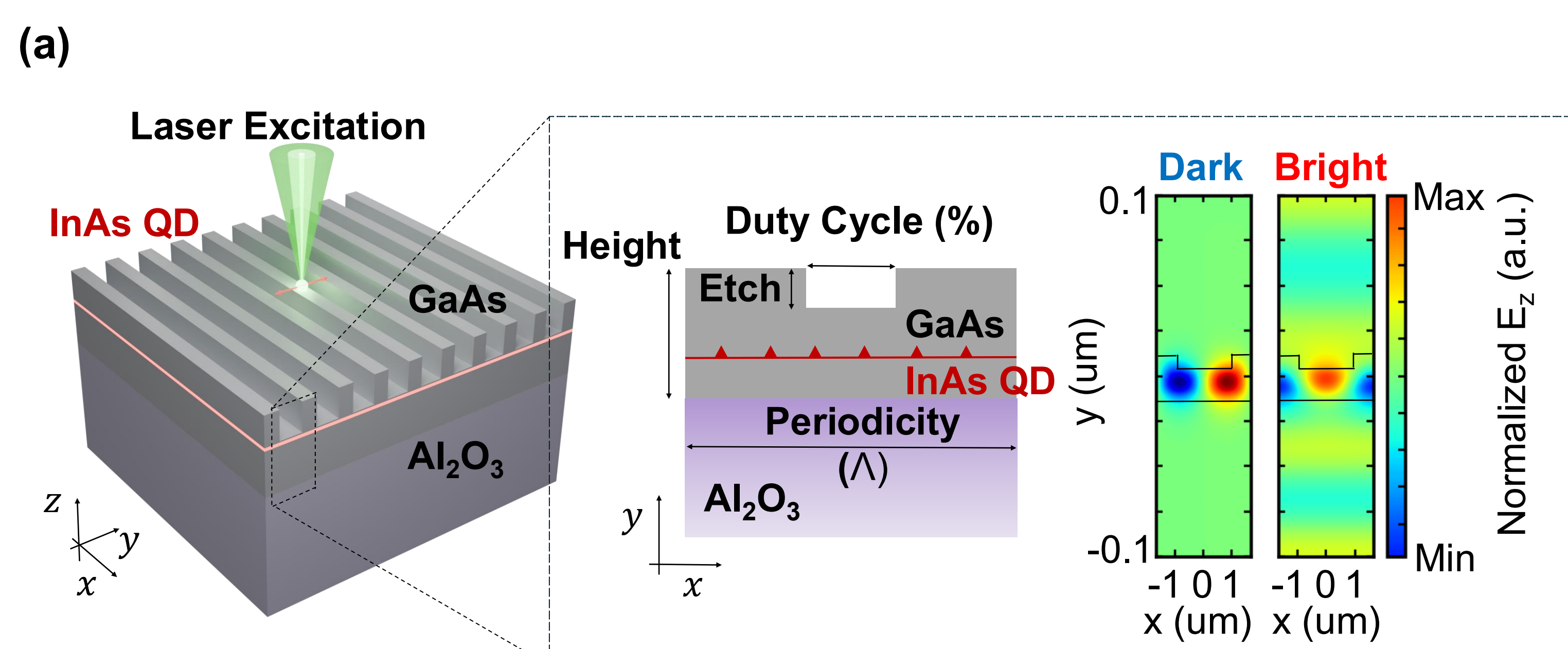
E_0 : midgap energy

v, δ : coefficients of k-p perturbation theory

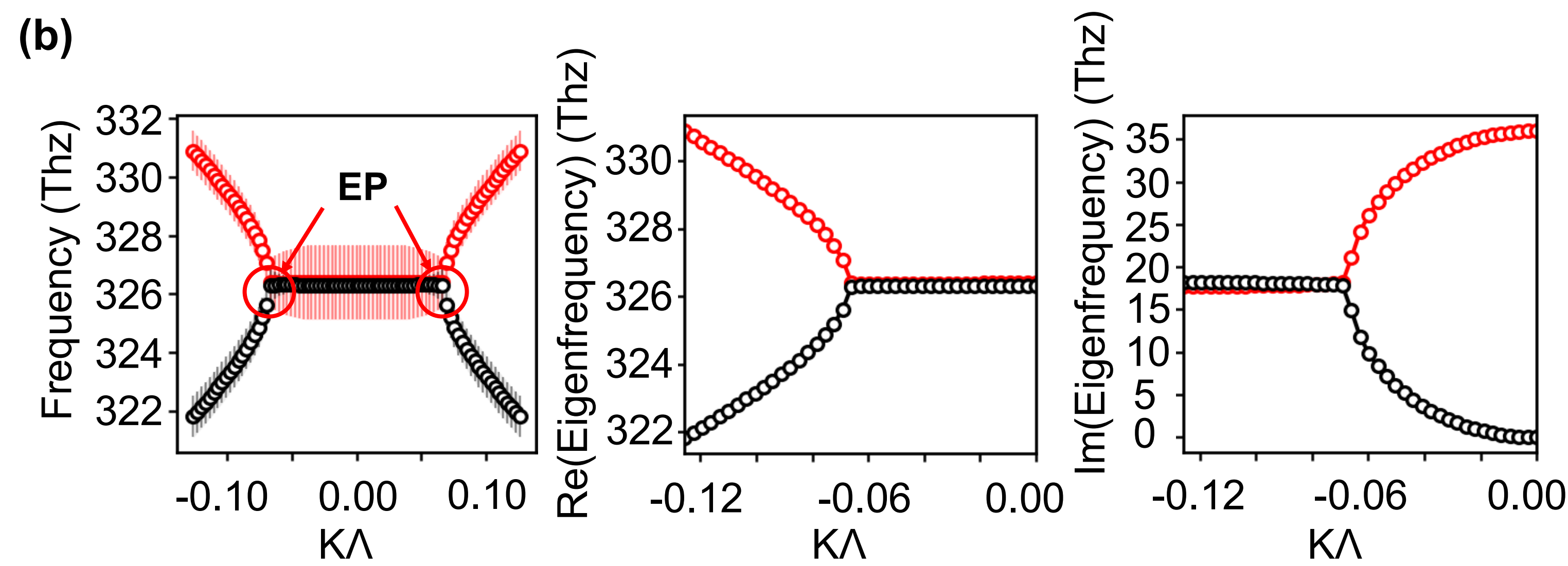
γ_{nr} : nonradiative loss of both modes

γ_b : radiative loss of the bright mode

Simulations

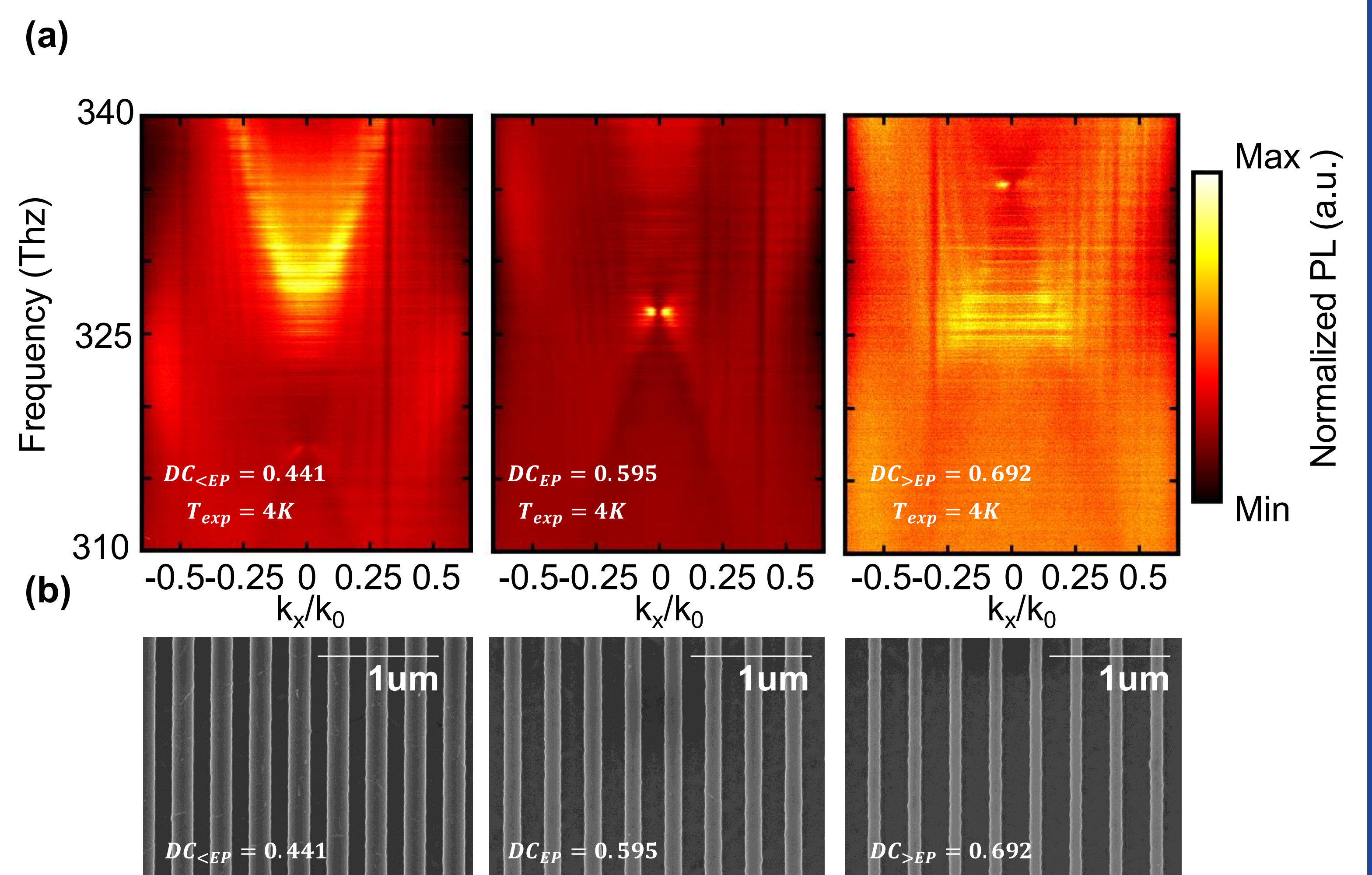


Schematic of the device and unit-cell geometry, and the simulated electric field profiles of the two eigenmodes, corresponding to the non-radiative dark mode (BIC mode) and the highly radiative bright mode.



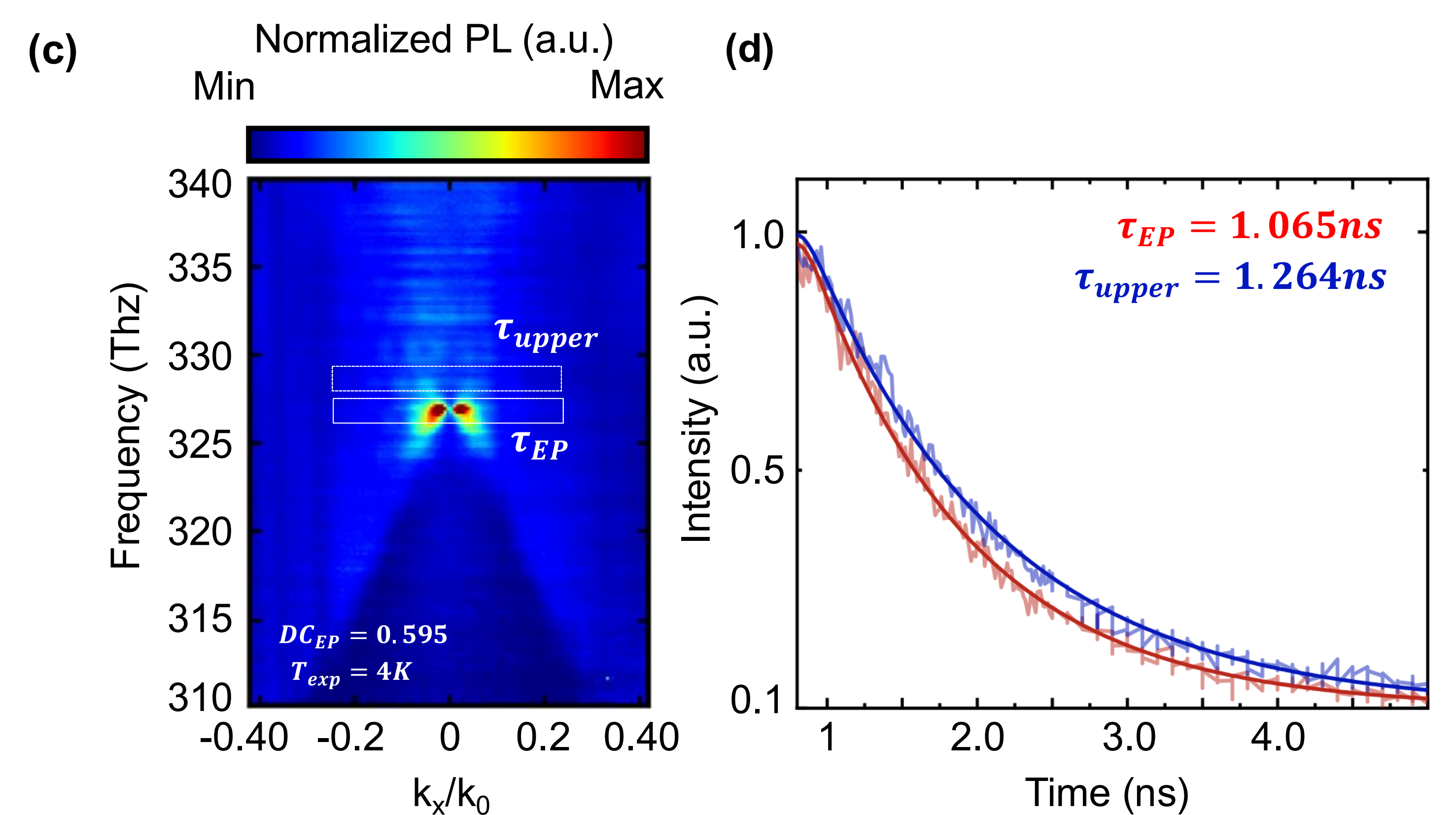
FEM-calculated band structure of the grating. The real parts of the eigenfrequencies (circles) exhibit band coalescence at the exceptional point, while the imaginary parts (line bars) split, indicating radiative losses.

Results



(a) Duty Cycle (DC) dependent evolution of the ARPL maps: below, at, and above the exceptional point condition DC_{EP} . As the duty cycle approaches DC_{EP} the two modes merge in momentum space, forming a characteristic signature of mode coalescence at the exceptional point.

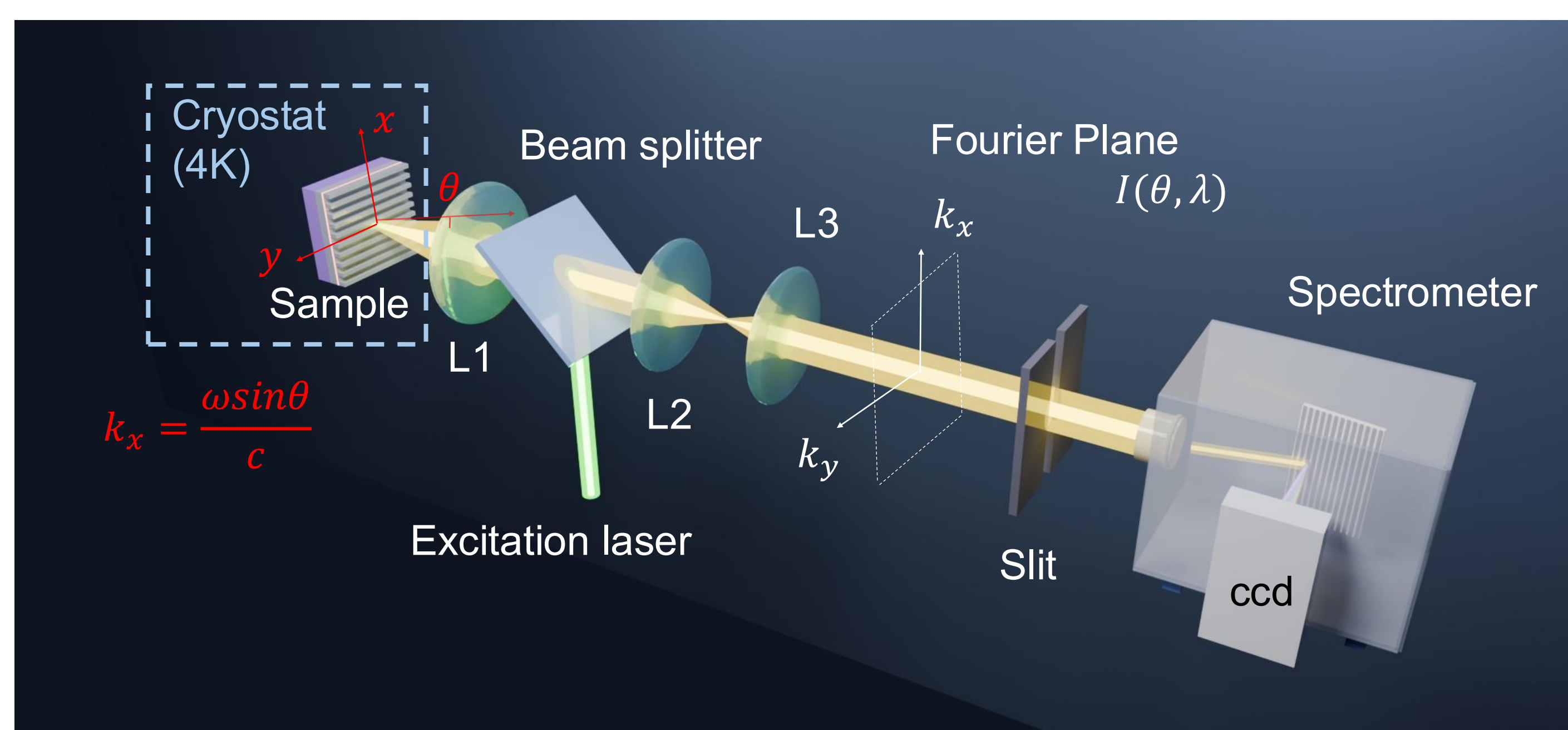
(b) Corresponding SEM images of the fabricated grating structures.



(c) ARPL map used to select the spectral regions corresponding to the EP and the upper band. Two localized emission spots are observed, the extracted momentum and frequency positions of these emission features are in good agreement with the band structure shown in Simulation Section.

(d) Time-resolved photoluminescence (TRPL) decay curves measured at the EP (red) and upper band (blue), showing a reduced lifetime at the EP $\tau_{EP} = 1.065$ ns compared to the upper band $\tau_{upper} = 1.264$ ns. The faster decay at the EP indicates an enhanced radiative recombination rate. (The theoretical LDOS enhancement factor of a passive EP is 4.)

Experiment



Angle-resolved photoluminescence (ARPL) spectroscopy setup. The sample is mounted in a 4K cryogenic system, allowing direct measurement of the emitted light as a function of in-plane wavevector k . The emitted light is mapped onto the Fourier plane, converting emission angle θ to in-plane wavevector $k_x = \omega \sin \theta / c$.

Conclusion

- We demonstrated **exceptional-point (EP) gratings** on an InAs/GaAs quantum dot membrane.
- The EP was formed with a **duty-cycle-engineered second-order grating** by degenerating radiative (indirective) and direct backward propagating waves.
- ARPL measurements confirm the bandgap closure at the EP duty cycle, showing the enhanced PL signals near the EP band.
- TRPL measurements demonstrate a reduced emission lifetime at the EP, enhancing **spontaneous emission via EP-induced LDOS** modification.
- Our work shows the potential for enhanced quantum light emission via EP gratings.

Reference

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- [2] A. Pick, *et al.* "General theory of spontaneous emission near exceptional points" *Opt. Express* 25(11) (2017)
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